A Novel Hybrid Model for Vendor Selection in a Supply Chain by Using Artificial Intelligence Techniques Case Study: Petroleum Companies

Mohsen Jafari Nodeh[1], M. Hanefi Calp[2](✉), and İsmail Şahin[3]

[1] Department of Informatics Institute, Gazi University, Ankara, Turkey
mohsen@gazi.edu.tr
[2] Department of Management Information Systems, Karadeniz Technical University, Trabzon, Turkey
mhcalp@ktu.edu.tr
[3] Department of Industrial Design Engineering, Gazi University, Ankara, Turkey
isahin@gazi.edu.tr

Abstract. Oil is an important strategic material which is associated with vital and major components of national security and economy of each country. In this context, supplier selection in a supply chain of oil companies has a direct effect on immunization and optimization of the production cycle, refining and distribution of petroleum, gas and petroleum products in oil producer and exporter countries. Therefore, creating and owning a purposeful and intelligent process for evaluation and analysis of suppliers is one of the inevitable needs and concerns of these countries. Many of the methods which are currently widely applied in the management of oil companies utilize traditional supplier selection methods which are unfortunately limited to individual and subjective evaluation in weighing decision maker's criteria, incorrect assessment rules, and inefficient decision-making methods. In this paper, with an in-depth look at the supplier selection in supply chain management of oil companies project, a novel model has been proposed based on an object-oriented framework. This model which finally leads to optimal selection and ranking of suppliers, reducing the time and cost in the selection process and also reduced human errors by using data mining techniques and neural networks in the reasoning method cycle based on the case. The proposed model was implemented on data bank information of the Oil Company. Finally, the results of the proposed model are compared with several other models. Results show that using reduced errors, improved accuracy, and efficiency the proposed model has been able to have a good performance in the supplier selection.

Keywords: Vendor selection · Supply chain · Artificial intelligence · Supplier selection

© Springer Nature Switzerland AG 2020
D. J. Hemanth and U. Kose (Eds.): ICAIAME 2019, LNDECT 43, pp. 226–251, 2020.
https://doi.org/10.1007/978-3-030-36178-5_19

1 Introduction

One of the main goals of data mining is finding models, patterns and repetitive rules through large amounts of data. In comparison with other branches of artificial intelligence, data mining significantly increases logical reasoning and learning of mathematical models. The results of Supplier Selection Based Data Mining represent a repeating pattern of "if … then …" and understanding and interpretation of the rules is very simple. The decision makers realize suppliers' problems by analyzing these rules. Common learning methods data include: Rough set, Genetic algorithms, Decision tree, neural network, case based reasoning (CBR).

Due to the product's uniqueness, frequency of purchase, and the ability to experience the historic decision of oil companies, this paper uses supplier selection method based on data mining techniques.

In the selection of suppliers using the same sample is very practical and has shown its effectiveness as a supplier selection strategy. In addition, the high volume of data with techniques of data mining is fully compatible with the spirit of the industry.

Due to the complexity of reasoning, CBR must solve several key problems:

- Case warehouse which is under the management and support of CBR, should resolve time and space and complexity problems.
- Character distilling and weight setting in CBR should create the possibility for comparing the current and main case existing in the memory.
- CBR must have a good appeal system power, the ability to maintain high reservoir and can support users to extract knowledge effectively.

1.1 Supplier Selection Methods

Scientific research to find a systematic approach in decision making, purchasing and supplier selection commerce is developed in more than three decades. In a review of methods for supplier selection by de Boer et al. [1] writers the trade Supplier Selection Process is divided into four stages. (1) Defining the problem, (2) defining the criteria, (3) preliminary evaluation of suppliers and (4) making a final choice [2]. Preliminary evaluation reduces "all" suppliers to a smaller set of acceptable 135 suppliers, ways of doing this step can be classified into four categories: Categorical [3], data envelopment analysis (DEA) [4–8], Clustering analysis (CA) [9] and case based reasoning method [10–13]. The suggested methods for the final selection stage are also classified in a linear weighting models [14], Total Cost of Ownership models [15] mathematical programming models [4, 5, 16], statistical models [1] and artificial intelligence models form [17–22]. Most of the suggested methods at this stage are dependent on mathematical programming and weighting models [10]. Mathematical Programming allows the decision maker to decide the issue using variables in the objective function to formulate for maximizing revenue and minimizing costs. Weber et al. [4] presented a multi-objective planning In order to determine the environmental characteristics of a number of conflicting provisions in the form of criteria for selecting providers. They also developed models using non-harmonized trading strategies to allow Transactions with suppliers selected by the multi-objective programming model. They showed that

the values of a chosen supplier alters based on the number of suppliers [2]. In linear weighting models, weight allocates to the criteria, the highest weight indicates the highest importance.

Degrees of criteria are multiplied by their weight and the sum is shown as a number for each supplier. Trade supplier with the highest total degree is suggested for selection. Lee et al. [14] provided the supplier select and management system which it uses a linear weighting model to calculate the weight of sensible and insensible criteria and categorizing the suppliers' performance. This system can identify the chosen supplier's weaknesses and compare them with other suppliers. The supplier select and management system provide ways to improve a trade supplier's performance [2].

Suppliers Analysis and Selection Techniques

In order to provide supplier selection processes, De Boer et al. [1] have done a wide examination on reported decision making approaches in literature from the time when Weber et al. [23] reviewed it. Unlike other reviews, De Boer et al. covered all stages of the selection process includes primary definition the formulation of criteria, preliminary evaluation and final selection of the best suppliers. Figure 1 has summarized some of these popular presented methods by De Boer et al. and conceptual studies after them including works of Das et al. [24], Kumar et al. [20, 22] and Chai et al. [25].

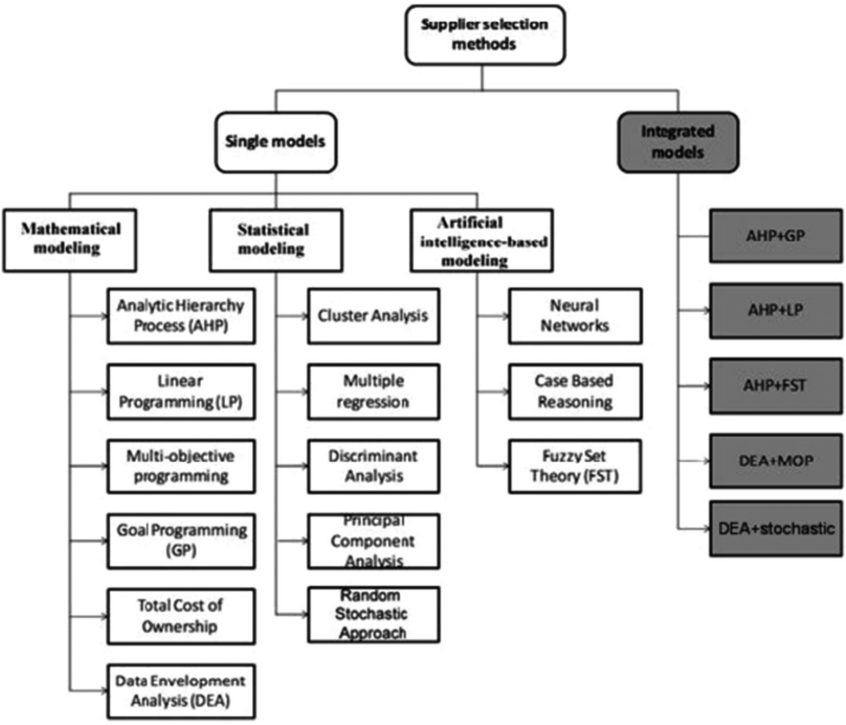

Fig. 1. Supplier selection methods.

Linear Weighting Patterns

This is a method that solves the single-source problems. It is described by a weight to each rule, whatever the weight the higher it is more important. The outcome of any supplier, summary score of every law and product of every law, is the supplier's weight. So every person who obtained the highest outcome will be the best supplier.

The Analytical Hierarchy Process (AHP) is the most widespread method of a linear weighting model [26]. This simple method is applicable and effective which it is used in choosing a supplier [14] and evaluating suppliers [27]. The problem with AHP is, it depends on experts and the role of elements and personal cases in it are more. In addition, sometimes using selected features will be incorrect and will cause diversion in the result of the decision.

Transaction Cost Approach (TCA)

Transaction cost approach (TCA) is a normal approach to solve single purchase problems. Its main idea is calculating purchase cost which includes selling price, purchase costs, transportation costs of a supplier that can meet the needs. Compare to different purchases of suppliers, the one who has the lowest price will be selected. Timmerman invented the cost rate approach in 1986 which calculates product's, transportation's and services' final price that are related to related price to select a supplier [3]. This method can calculate the certain ration of each law in the final price. Roodhoof et al. [28] followed the activity method based on value (Activity based on casting) or ABC for choosing suppliers in 1996. Monczka et al. [29] in 1998 suggested that the supplier method in oil companies chooses using direct and indirect prices and analyzing these prices in activity purchases. Degraeve et al. [30] before 2000, compared 13 selected patterns and calculated the price of each pattern with real purchase data of the Belgian steel company and the consequence is that mathematical programming approach is better than the evaluation approach and multi-products are better than singe products.

Mathematical Programming Method

Mathematical programming method is an important method to solve problems in one or more sources, including programs for single or multi product, linear and nonlinear Programs combined. Since the program is easily explained with mathematical methods and software Supplier selection is a good way to solve problems. In 1974 Gaballa for the first time used the linear system for his problems [31]. Benton set A non-linear program features with the ability to select multi-product, multi-source, multi-client, limited supply, limited amount, Storage costs and the order of prices for its purpose [23]. Ghodsypour and O'Brien invented a combined non-linear pattern which solves multi sources with the ability to limit selection and determines the number of purchases. The aim of this project is to reduce and minimize the cost of buying, Store, transfer and order that the quality and limit its production capacity to itself [32]. According to the purpose of minimizing costs, the single-purpose model doesn't have the ability to solve multi-rule problems. The solution is the multi-purpose linear programming model that can solve the supplier selection criteria challenges. In 1993, Weber & Current used a multi-functional linear model for supplier selection, the price, quality and supplying them as their goals which can choose to monopolize demands, poli tics, and the chosen amount [33]. It should be noted that there is no sufficient

scientific and historical data about the complexities of society, and production systems and therefore it is difficult to have problems to abstract mathematical models and precise methods of mathematics for solving it. In addition, to obtain an analytical solution, it is very difficult when problems are large and complex [34].

Traditional selection methods based on analysis of statistics from suppliers which are at high levels have major flaws which is mentioned in the following:

- Decision-making which Lacks a dynamic status and learning capability, remains suspended in the nature of these patterns. The range is specified and its adaptability is poor. So we can't function effectively with inaccurate information [35].
- The decision making standard is based on people's taste and arranging lists and weight does not matter.
- The result for supplier selection is shown as a score or a degree that it is not easy to understand. Moreover, it doesn't define the quality of the supplier's operation. So, monitoring the supplier for improvement is difficult [34].

Since the results of this choice cannot be explained, finding hidden rules and its covered patterns is hard. Thus, decision making results are not predictable. There are two ways: from one hand a large number of mental patterns exist and on the other, is the action taken on a limited number of them. So it is necessary to invent ways which with their powerful transmission, decrease the gap between theory and reality.

CBR Approaches for Suppliers Selection Based on Data Mining

CBR is a set of three thinking-logical, innovation and inner sense of human beings. In 1982, Schank writes a book named dynamic memory which was the start of recent studies about the details of the case-based reasoning method and presented the CBR creation method in computers [36].

In some ways, data mining and CBR technology and results, both used to make decisions, and both began Similar. In the CBR system, a lot of machines and information exists such as: The structure and content of the item, examining the similarities, survival of the item Storage, review, the list's features, item Points and etc. Also there are some problems to obtain these information [37].

In 2006, Li and Shiu invented the information classification system with CBR [38]. In 2007, Huang & Chen made a personal network learning system based on hereditary algorithm with CBR [39]. In recent years, Liu Choy and Lo used CBR in supplier selection and have done extensive research in this field [13]. Chan and Harris have done a research on the environmental management of supplier performance evaluation [11]. In 2001, Kim and Han suggested the combined data mining and case-based reasoning methods based on support vector machine (SVM) in order to predict the classification of bonds. This research Showed that the combination of CBR and SVM, is an effective method in transition index and can increase the accuracy of prediction significantly [40]. In 2011, Zhao et al. [34] presented a new model using CBR based on clustering data in choosing the suppliers of China's National Petroleum Corporation. In the context of using and combining data mining methods with the case-based reasoning method.

It is important to note that these studies actually suggest or provide a systematic model to improve the ability of trust management decisions. Thanks to advances in

science, particularly statistics, nowadays researchers have more facilities to build more reliable models. In this paper, the novel intelligent model to predict the optimum supply achieving these objectives is presented.

2 Material and Method

2.1 The CRISPs-DM Process

This process includes the environmental data monitoring and analysis functions. In the DM-CRISP model, three steps are taken to monitor and evaluate the data. These steps are Business Understanding, Data Understanding (Analyzing and examining data) and Data Preparation. In business understanding phase, since the scope of the reviewed activities is related to project supply chain management and in particular the project supply management, the basic familiarity was studied with the supply chain and processes of providing project management was taken place. In data understanding phase, data mining process requires some understanding of the data that it's investigating. This process has got two important functions such as "Collecting initial data" and "Explaining and describing data". In final phase, data was prepared to process data mining [41, 42].

In this research, 1430 records were collected which after data cleaning was reduced to 1118 records. To overcome these problems, various methods and solutions were used which some of them will be mentioned in the following. After preparing the data, 36 attributes for each item were obtained and finally 29 attributes were selected which can be seen in Table 1.

Table 1. Proposed model attribute names.

No	Attribute	No	Attribute	No	Attribute
1	Project name	11	Responsiveness and accountability	21	The number of payment in contract
2	Order name	12	Country code	22	Presentation and answering capabilities in meetings
3	Supplier name	13	Time objectives	23	The company AVL
4	Supplier's nationality	14	Flexibility in production	24	Percentage of payment
5	Producer or dealer	15	The number of proposals for contract	25	Total score in the technical criteria
6	Price forecast	16	Cooperation in specific technical demands	26	The main part of the contract
7	The suggested retail price	17	Company status near the employer	27	Number of different products presented in the proposal
8	Ability to write and recommend proposal	18	The original offer of contract	28	The main part of the contract payment
9	Delivery time	19	Communications	29	Moon and year of proposal
10	Delivery type	20	Win or reject		

2.2 The Proposed CBR Framework

This framework is the operational environment is the case-based reasoning (CBR). Operating modules that put the 4R process in the CBR cycle are embedded in this structure. Embedded triple module and figure of the model (Fig. 2) are as follows:

- CBR case development module
- CBR solution provide module
- CBR advisor module

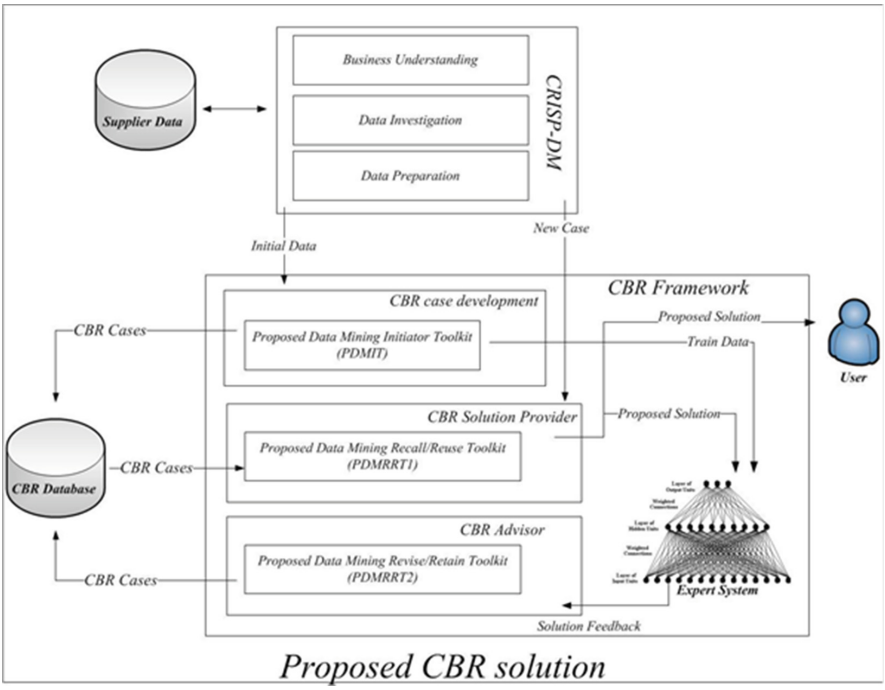

Fig. 2. Proposed model architecture

CBR Case Development Module

Theoretical Assumptions

It is assumed that the primary training base cases an n member set of our suppliers: Supplier Initiation Matrix:

$$SIM = \begin{bmatrix} SI_1 \\ \vdots \\ SI_n \end{bmatrix}; SI_i: i^{th}\ Supplier\ for\ case\ initiation\ i = 1 \ldots n$$

It is assumed that we have the following after passing by the CRISP-DM modules containing m descriptive variables from the suppliers:

Attributes Initiation Matrix:

$$AIM = \begin{bmatrix} AI_1 \\ \vdots \\ AI_m \end{bmatrix}; AI_j{:}\, j^{th}\ Attributes\ for\ case\ initiation\ j = 1\ldots m$$

Using the SAIM Matrix described as follows, a relationship is placed between the suppliers and the descriptive variables:

$$\text{Supplier/Attribute Initiation Matrix}: SAIM = \begin{bmatrix} SAI_{11} & \ldots & SAI_{1m} \\ \vdots & \ddots & \vdots \\ SAI_{n1} & \ldots & SAI_{nm} \end{bmatrix};$$

$$SAI_{ij}{:}\ the\ decriptor\ for\ j^{th}\ attribute\ of\ i^{th}\ Supplier\ in\ initiation\ \ i = 1\ldots n,\ j = 1\ldots m$$

According to the above definition we have:

$$\text{SIM} = \text{SAIM} \S \text{AIM}$$

Which § in it is as in which the elements of the matrix and describing them next to each other. The present assumption is that after the transition from modules DM-CRISP, in the decision variable p module earns about suppliers be:

$$\text{Decision Matrix}: DM = \begin{bmatrix} D_1 \\ \vdots \\ D_p \end{bmatrix}; D_l{:}\, l^{th}\ Decision\ Value\ for\ l = 1\ldots p$$

Using SDM as follows matrix varies between suppliers and decision-making about suppliers' relationship is in place:

$$\text{Supplier/Decision Matrix}: SDM = \begin{bmatrix} SD_{11} & \ldots & SD_{1p} \\ \vdots & \ddots & \vdots \\ SD_{n1} & \ldots & SD_{np} \end{bmatrix};$$

$$SD_{il}{:}\ the\ decriptor\ for\ l^{th}\ decision\ of\ i^{th}\ Supplier\ \ i = 1\ldots n,\ l = 1\ldots p$$

According to the above definition we have:

$$\text{SIM} = \text{SDM} \S \text{DM}$$

Which § in it is as in which the elements of the matrix and describing them next to each other.

If all the information about the suppliers in terms of descriptive variables and decision variables are considered, we have:

$$\mathrm{SIM} = \mathrm{SM} \S \mathrm{M}$$

Which in it:

$$\mathrm{SM} = \left\{ \begin{array}{ccccccc} SAI_{11} & \dots & SAI_{1m} & SD_{11} & \dots & SD_{1p} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ SAI_{n1} & \dots & SAI_{nm} & SD_{n1} & \dots & SD_{np} \end{array} \right\}$$

$$M = \left\{ \begin{array}{c} AI_1 \\ \vdots \\ AI_m \\ \hline D_1 \\ \vdots \\ D_p \end{array} \right\};$$

Implementation of the Module

As also shown in Fig. 2, the module in addition to the training data set for training the neural network as an expert, is responsible to develop the case base for the proposed model. Before taking any action necessary to the importance of each aspect of which is specified in the data base. For this purpose, the technique gain ratio is used. The weights obtained in this way is used for the extraction of decision variables 8 variables that have the most weight as a decision variable and the remaining features are considered as an explanatory variable.

Also, as indicated in Fig. 3 entire database should be three sets of training data, test data collection and validation data set are divided. For this purpose, 70% of data as training data, test data and 20% as the remaining 10% were considered as validation data set. As mentioned, the other measures contained in the training data set for neural network training modules are created as connoisseurs. This dataset contains data on the educational database for all descriptive characteristics and decision. Also, the training data set of decision variables used to train the decision tree mining rules. Mining rules have been used to produce case base.

Proposed Data Mining Initiator Toolkit (PDMIT)

Data mining techniques used in the PDMI toolkit described early in the development of the case base:

– *Weighting using Gain ratio*

Since the database provided by the suppliers include numerical variables, class variables of interest rate method for weighting characteristics is used. How to calculate interest rates are as follows:

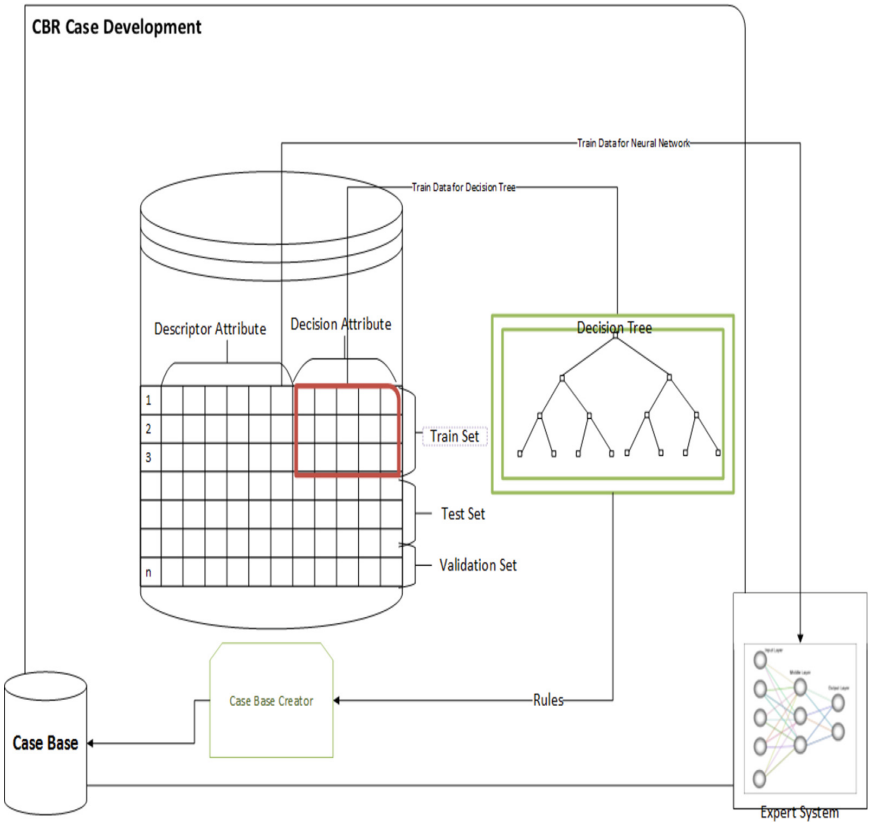

Fig. 3. CBR Case development module architecture

Step 1: searching for the means of the main data. Since the characteristics of each feature on the original data are different, so first of all we mean the quality of each index declining oil companies' equation is computed. (After obtaining the means list of each variable). Finally, the degree and significance of each variable is categorized based on score producer. Calculating the singular coefficient: $Y_{in} = a_i X_{in}^{bi}$ which i is an integer, according to a numerical value to each of which we look back, bi and ai are decline or recursive parameters. N means the number of hosts. x_{in} means the value of any feature that is provided by the Department of Management evaluation.

Calculating b_i and a_i: $Y_{imax} = a_i X_{imax} b_i = 1.30$. $Y_{imax} = a_i X_{imax} b_i = 1.30$

These are based on the valuation of regular management department.

$$a_i = 1.3/power(X_{imax}, b_i) b_i = ln(1.3)/ln(X_{imax}/X_{imin})$$

Step 2: Calculate information gain: based on the ID3 decision tree, we can use instead of the classification decision of A (instead of obtaining information).

Obtaining information as mentioned function is defined as follows: we assume that A has different values different to v number. We can use A to divide S to v number of

sets: $\{s_1, s_2, \ldots, s_v\}$. We can use A to divide S to a subset of v s. If A is selected as a test example (which covers the clearest example). So those subunits indicate the expanded and separate branches of S. we assume that S_{ij} is the record number of group C_i in the subcategory of S_j. So the intended entropy or information is ramified from A:

$$E(A) = \sum_{j=1}^{r} \frac{S_{ij} + \cdots + S_{mj}}{S} I\left(S_{ij} + \cdots + S_{mj}\right)$$

The information that will be achieved in branch A is as follows:

$$Gain(A) = Info(s_1, s_2, \ldots, s_m) - E(A)$$

After selecting the properties influencing the choice of supplier, these characteristics need to be assigned weights and the weights are calculated in a way to reflect the effect of each of the following characteristics. Since the Selected features that include batch and numerical variables are variables, and on the other hand because the distance calculation in CBR technique requires weight values that are in the range of zero to one, the gain ratio technique in the orange Canvas software is used. Also, the more of these rhythms will be used for the selection of decision variables in the proposed framework. Figure 4 shows the weights obtained ReliefF techniques and the Gain ratio.

	Attribute	#	ReliefF	Gain ratio
1	No. of suggesti...	3	0.037	0.070
2	TOTAL SCORE	C	-0.001	0.059
3	Fulfillment / Technical Spec Compliance	C	-0.009	0.056
4	delivery location	C	-0.008	0.055
5	status code	8	0.069	0.050
6	No. of peyment step	5	0.008	0.047
7	delivery type	C	0.097	0.046
8	Writing ability & Appearance	C	-0.005	0.038
9	main suggestion	C	-0.010	0.035
10	Proposal Quality / Preciseness	C	-0.008	0.035
11	AVL code	3	-0.030	0.034
12	Flexibility	C	-0.003	0.029
13	(actual/ estima...	C	0.001	0.020
14	Presentation / Meeting	C	-0.036	0.020
15	Communication	C	0.004	0.019
16	Response	C	-0.020	0.018
17	Total delivery time (day)	C	0.002	0.009
18	actual price ()	C	-0.008	0.007
19	supplier code	C	-0.025	0.004
20	country code	C	-0.032	0.004
21	project code	5	-0.018	0.002
22	DEVIATION No.	C	0.008	0.002
23	proposal year	C	0.002	0.001
24	Pre Peyment	C	-0.021	0.001

Fig. 4. Weighting attributes by ReliefF and Gain ratio.

– *Training neural networks as the expert*

In this paper, Multilayer Perceptron Neural Network (MLP) is used as the expert. For training this network, all features including the training of the decision variables and descriptive variables are used. Therefore, the input layer of the network will have 29 neurons, hidden layer also has 20 neurons and also the output layer contains a neuron that identifies the input supplier will be selected or not selected. Parameters set for MLP neural network training are as follows:

- Number of repeats: 500 times
- Learning rate: 0.1
- Activation function: Sigmoid

MLP neural network has been trained in the testing phase with new examples. In addition to its opinion to select or deselect suppliers based on thresholds defined for the decision variables, sends its feedback to the CBR advisor module till it improves the performance of the database. This expert opinion is applied for every test. In the proposed model validation phase, the expert does not interfere with the operation of this phase. In the proposed model validation phase, the expert does not interfere with the operation of this phase.

MLP neural network implementation is done as an expert in MATLAB with the write function as MLPexpert.ml. As indicated in Fig. 5 the number of neurons in the input layer is 29 neurons that includes all the decision variables, and descriptive variables problem; Moreover, 20 neurons are in the hidden layer and a layer neurons output indicates the supplier is selected or not selected.

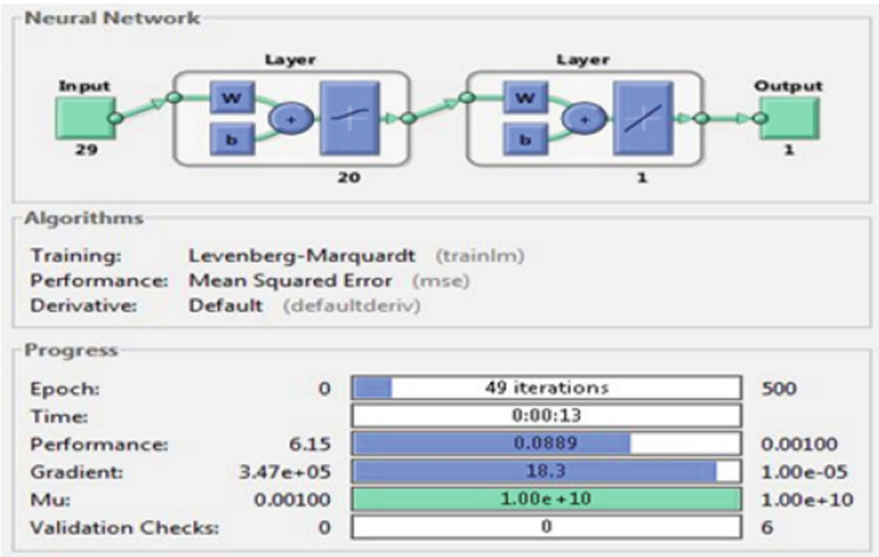

Fig. 5. Training MLP in MATLAB.

– *Establishing a case-base based on extracted rules from the decision-tree*

The main objective of the module of creating and developing cases, is providing a case base based on rules. Decision Tree algorithm rules by using the training data set and only decision variables are done. The output of the decision tree will be if-then rules that the foremost step includes thresholds for 8 variables and the step after is the determination of selection and/or skipping suppliers. After extracting data, in the case base creator, these rules are used and all cases that approve at least one of these rules have been placed in the database. In other words, each instance of database values in their decision variables, which are accepted by one of the extracted rules, are placed in the intended case-base. The noteworthy point is that in establishing case-base, only existing samples in the training data set are used and sample testing phase of the neural network as the expert are added or revised.

In the PDMIT Module, the numbers of features as decision variables were used to train the decision tree mining rules. To identify and select these characteristics of the weights obtained from the section above and 8 features the highest weights are used as decision variables selected. In other words, the first 8 features as decision variables are considered in Fig. 4.

With regard to database mining and decision variables related to suppliers, decision tree training data set can be created in such a way that only the relevant part of the decision variables from the database. In other words, the dataset used to train the decision tree model training data set includes all samples and only features of the proposed decision variables have been identified. As shown in Figs. 6 and 7, Clementine SPSS software and Orange has been used to train the decision tree. Decision tree rules if-then taught two classes for each supplier is selected and not selected on the basis of the decision variables creates.

CBR Solution Provider Module

Theoretical Assumptions

It is assumed that a new case to assess and provide the desired variables by the proposed framework:

New Case Matrix: NCM

It is assumed that after the passage of the modules of the DM-CRISP with m variable in our NCM:

$$\text{Attributes Matrix}: AIM = \begin{bmatrix} AI_1 \\ \vdots \\ AI_m \end{bmatrix}; AI_j: j^{th}\ Attributes\, for\, case\, j = 1 \ldots m$$

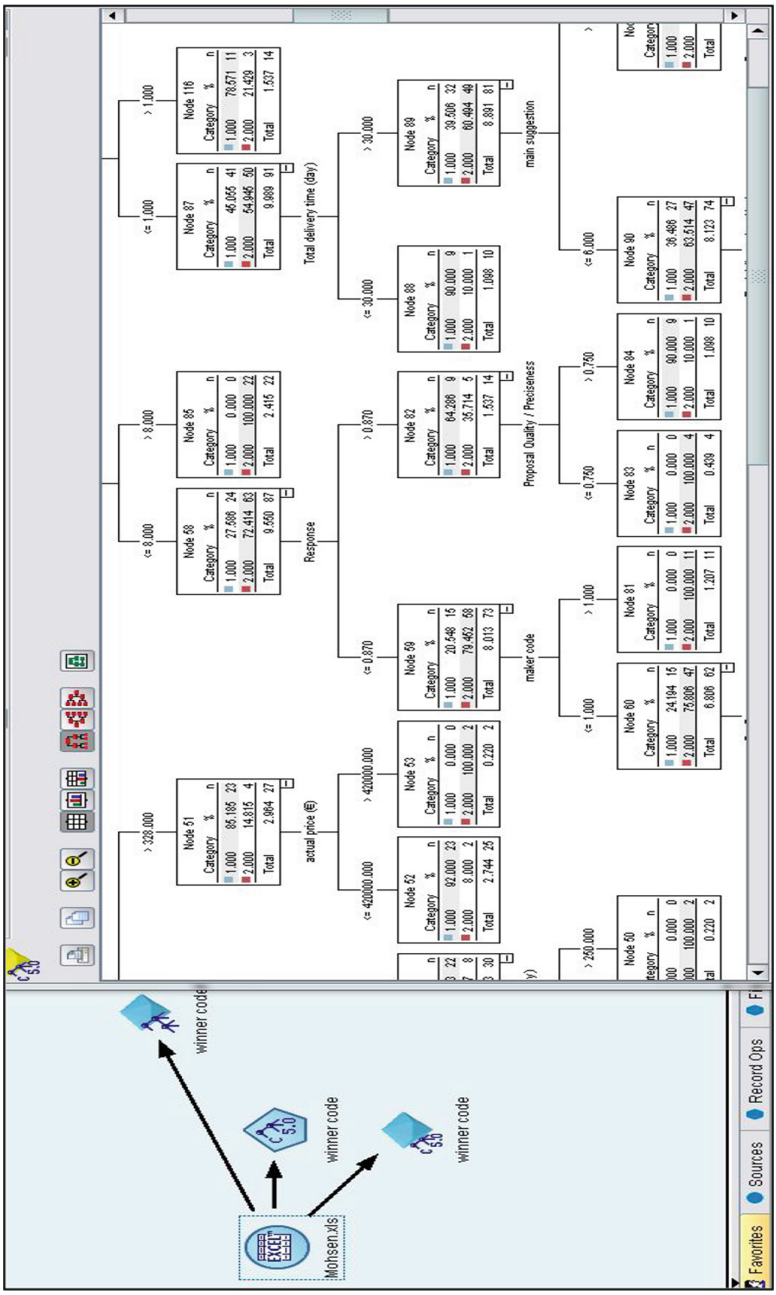

Fig. 6. Training decision tree.

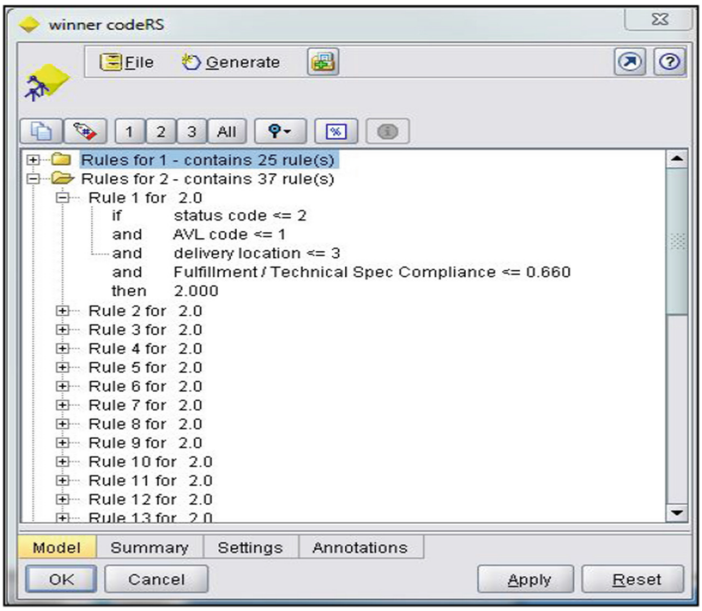

Fig. 7. Extracted rules from SPSS.

Using the Matrix CAM, the following of the new case and descriptive variables related to that relationship:

$$\text{Case/Attribute Matrix: } \mathrm{CA}M = [\,CAM_{11} \quad \ldots \quad CAM_{1m}\,];$$

$$CAM_{1j}\text{: the decriptor for } j^{th} \text{ attribute of New Case}, j = 1\ldots m$$

According to the above equations:

$$\mathrm{NCM} = \mathrm{CAM} \S \mathrm{AIM}$$

Which § in it is as the multiplier of elements of such Matrix and descriptive next to each other.

Implementation of the CBR Solution Provider Module

After creating the database from cases in the case development module, the model with new examples to follow is the nearest case to the new sample in case-base. Case solution strategy proposed new case will be retrieved. So, what this module will be most interested in how to find the nearest is using a new case. Since the cased in the database include the numerical variables and the categorical parameters, SOM clustering algorithm was used based on the Prototype-K for clustering databases and improve the efficiency of the recovery. After clustering, retrieving the nearest case in search for a cluster that its center has the least distance to the new sample, is enough.

Table 2. Samples of extracted rules from decision tree.

No	If	then
1	if status code <= 2 and AVL code <= 1 and delivery location <= 3 and Fulfillment / Technical Spec Compliance > 0.660 and supplier code <= 328 and country code <= 7 and No. of payment step <= 1 and Total delivery time (day) <= 72 and (actual/ estimate) <= 0.986	1
2	if status code <= 2 and AVL code <= 1 and delivery location <= 3 and Fulfillment / Technical Spec Compliance > 0.660 and supplier code > 328 and actual price (€) <= 420,000	1
3	if status code > 2 and Fulfillment / Technical Spec Compliance >0.875 and AVL code <= 1	1
4	if status code <= 2 and AVL code > 1 and (actual/ estimate) <= 1.022 and Proposal Quality / Preciseness <= 0.820 and DEVIATION No. <= 8 and Response <= 0.870 and maker code <= 1 and Pre Payment<= 70 and Flexibility <= 0.300	1
5	if status code <= 2 and AVL code <= 1 and delivery location <= 3 and Fulfillment / Technical Spec Compliance <= 0.660	2
6	if status code <= 2 & AVL code <= 1 & delivery location > 3	2
7	if status code <= 2 and AVL code > 1 and (actual/ estimate) <= 1.022 and Proposal Quality / Preciseness <= 0.820 and DEVIATION No. > 8	2
8	if status code <= 2 and AVL code <= 1 and delivery location <= 3 and Fulfillment / Technical Spec Compliance > 0.660 and supplier code >328 and actual price (€) > 420,000	2

Proposed Data Mining Recall/Reuse Toolkit (PDMRRT1)

The SOM Algorithm Based on K-Prototype

The SOM technique which was presented by Kohonen is one of the most distinctive unsupervised neural network models for applying clustering. A notable feature is that maintaining SOME similarities between input data and generate a map of the High-dimensional input, they largely on a two-dimensional space usually reflects as much as possible while trying to maintain the topological relationship between the input data. Similar to the same input data nodes on the map assigned. The SOM technique have been impaired in calculating the hands variables are similarities in the face of data; In this thesis to overcome this defect, the scale used for similarities in the techniques used Prototype-K. As stated in the second quarter of this technique by taking a mixture that includes both data and numerical variables are variables hands, the scale used to measure a representative sample. SOME proposed model-based algorithm K-Prototype are as follows:

1. All weights and learning rate α and neighborhood radius are initialized.
2. An input x of the input data set is presented to the network.

3. The winner neurons are selected based on criteria (The scale used in the Prototype-K technique):

$$d(X,Y) = \sum_{j=1}^{p} (X_i - Y_j)^2 + \gamma \sum_{j=p+1}^{m} \delta(X_i, Y_j)$$

4. Updating the winning neuron weight and neighboring neurons from k repetition to k + 1:

   For numerical variables:

$$\mathrm{w_{ij}(k+1)} = \begin{cases} \mathrm{w_{ij}(k)} + \propto \mathrm{(k)}\left[\mathrm{x} - \mathrm{w_{ij}(k)}\right] & \text{if } (\mathrm{i,j}) \in \mathrm{N_c}(k) \\ \mathrm{w_{ij}(k)} & \text{otherwise} \end{cases}$$

   For categorical variables:

$$w_{ij}(k+1) = \underset{j}{\mathrm{mode}}([x, w_{i,j}(k)])$$

5. Repeating steps 2 to 4 for the whole input vectors.
6. Reducing the learning rate and neighborhood radius determined based on an approach for the next period.

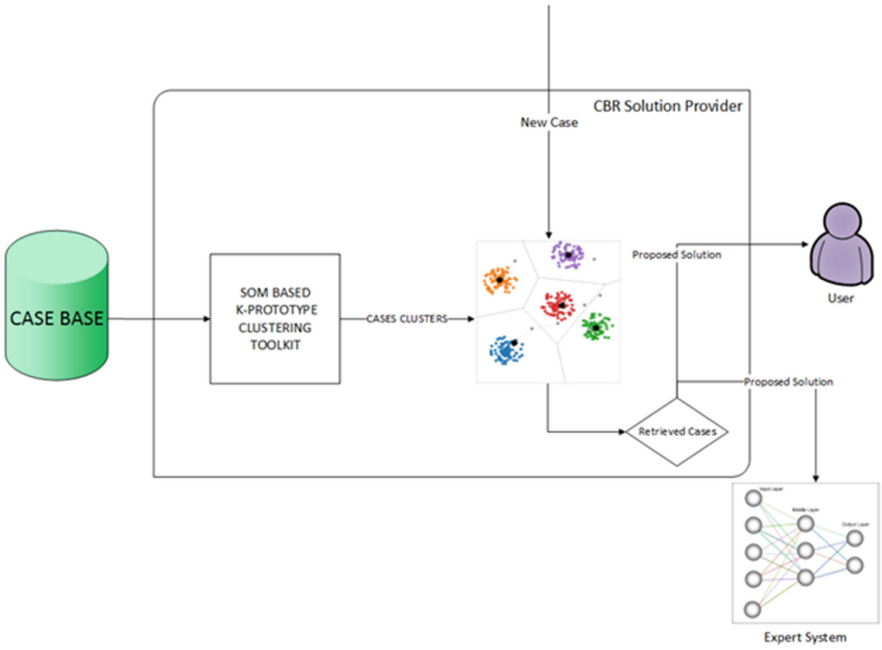

Fig. 8. CBR solution provider module architecture.

Repeating steps 2 to 6 until time realization termination condition (usually a certain number of repetitions).

Database Clustering

In the PDMRRT module, to improve the retrieval performance, nearest to the version of the base case, the SOM-based clustering technique is used K-Prototype. By implementing these techniques, the CBR database is partitioned into K clusters in such a way that the samples are similar to each other in a cluster. Davies – Bouldin index is used to calculate the optimal K that will be explained later.

Calculating the Davies-Bouldin Index

Clustering of clusters is aimed to achieve the highest minimum distance from each other and are within their cluster. In other words, clustering samples must be such that there is the highest concentration in each cluster and the cluster centers farthest from each other. Davis-Bouldin criteria for evaluating clustering is used. This index is a measure of clustering is that the compression and resolution into consideration and is calculated as follows:

$$index = \frac{1}{k}\sum_{i=1}^{k}\max_{i\neq j}\left(\frac{a_i + a_j}{d\left(c_i, c_j\right)}\right)$$

Where k is the number of clusters. a_i is distance Inside cluster correspond to cluster i. $d\left(c_i, c_j\right)$ equals to the distance between cluster i and j. Clustering is more appropriate to have the lowest index. To select the optimal k, for every k = 2, 3,…, 12 in the MATLAB software, the Davies-Bouldin index is calculated in a function called Davies-Bouldin.m which in Table 3, its results are shown. If specified in the table for k = 3 index is the lowest.

Table 3. Davies-Bouldin index for variant k.

K	2	3	4	5	6	7	8	9	10	11	12
Index	0.5203	0.3525	0.5135	0.5122	0.5199	0.5065	0.5134	0.5530	0.5339	0.5300	0.5346

Clustering Using SOM Based on K-Prototype

As previously mentioned, samples in the database are also characterized by a set of numerical and categorical characteristics; Therefore, it is necessary to choose K in parts of the algorithm SOM such as primary centers, distance calculation and update centers, this point should be considered. Centers for initialization vectors, the number of K samples of the base case selected randomly and were considered as centers of K clusters. Thus, even the values assigned to the properties of the cluster centers will be valid values. In calculating the distance as well as the data together with numerical variables and parameters are set. The purpose of using the software MATLAB function as distance.m was written to calculate this distance. SOM algorithm to calculate the distance in samples of cluster centers to cluster sample allocation is closer than the scale used.

Update centers separate functions to update numerical variables and class variables were written cluster centers. Numericupdate.m function has the task of updating numerical variables; this algorithm update function in SOM is standard practice. Nominal update.m function also updates the variables hands and with the addition of each cluster to cluster presented its fashion value and the variable batch assignment of the cluster center. After the implementation of SCM based on K- Prototype, for clustering on the base created in PDMIT module has been implemented. This implementation is done using MATLAB software and by function as CBclustering.m.

Since the cluster members during the testing phase CBR model by adding new models to the database of the proposed changes. It is necessary that in this phase update cluster centers are done using numericupdate.m functions and nominalupdate.m. Table 4 shows the number of each cluster after finishing the test phase.

Table 4. The number of each cluster after finishing the test phase.

Cluster	First	Second	Third
Number	212	134	416

In retain and reuse process, after identifying the nearest recycle the closest cluster, a new sample interval is calculated with all members of the cluster. Table 5 shows an example of the similarity of arranged new cluster members selected.

Table 5. Similarity between new case.

Index	78	91	105	371	42	8	296	511
Similarity	79.5%	74.12%	64%	48.23%	43.59%	41.82%	35.16%	32.91%

After clustering the cases to the nearest retain to new cases it is sufficient to search only the center of the cluster is the smallest distance using a new sample by reaching the nearest solution as the solution proposed by the expert presented.

CBR Advisor Module

Theoretical Assumptions

It is assumed that when a new case, with the last part of the NCM definition provided for evaluation and decision variables are analyzed by the proposed framework, having regard to the transition from modules DM-CRISP using AIM and CAM matrix for describing variable m describing the matrix NDM proposed as a solution is proposed. The CAM Matrix is provided as input to the expert system and the feedback returned by the expert system will be in the form of a matrix of variables decision which it is called PDM, we'll have:

$$\text{Proposed Decision Matrix: } PDM = [\,PDM_{11} \quad \ldots \quad PDM_{1p}\,];$$

$$PDM_{1l}: the\ decriptor\ for\ l^{th}\ decision\ \ l = 1\ldots p$$

According to the above equations:

$$\text{SIM} = \text{PDM} \S \text{DM}$$

Which in it, § is as the multiplication of such Matrix values and their description is beside each other.

The proposed solution module creator CBR strategy that was called NDM was studied next to the PDM Matrix and based on the Performance evaluation similarities of the cases in the database, update rate is determined. If you need to add new records to the CBR database, the new case will be added by this module. Finding similar operations using the structure of object-oriented cases occurs. Embedded behaviors in cases on receipt of decision variables in the valuation matrix of NDM and PDM, uses data mining to investigate the finding similarities in the result of the matrix of NDM deals and based on this modification of cases and added new cases or removing the existing case occurs. The action mechanism is like this that the output values of expert system is investigated and compared in a PDM Matrix form with an output value of NDM Matrix. If the variables are the same aim in both matrixes, variables are evaluated based on different thresholds for each decision, if the specified limit difference is less than framework threshold for each decision variable, the creation module output of the CBR solution is confirmed and is established in the case database and changes in the existing cases structure and consequently related to its laws. In contrast, if the difference between decision variables in NDM and PDM matrix outputs are greater than the specified threshold for each of the variables decision, Output module, causing the solution is reviewed. In this case, the module of the existing amendments to the CBR corrects and modifies the existing cases. For this purpose, the following process is done.

The new training case is again considered and creating platform cases of n + 1 sets of suppliers are also considered:

Supplier Revised Matrix:

$$SIM = \begin{bmatrix} SI_1 \\ \vdots \\ SI_{n+1} \end{bmatrix}; SI_i: i^{th}\ Supplier\ for\ case\ initiation\ \ i = 1\ldots n+1$$

In this case, the supplier has the same definition for m describes the former:

Attributes Initiation Matrix:

$$AIM = \begin{bmatrix} AI_1 \\ \vdots \\ AI_m \end{bmatrix}; AI_j: j^{th}\ Attributes\ for\ case\ initiation\ \ j = 1\ldots m$$

The SARM Matrix can be stated as follows between suppliers and descriptive variables:

$$\text{Supplier/Attribute Revise Matrix: } SARM = \begin{bmatrix} SAI_{11} & \dots & SAI_{1m} \\ \vdots & \ddots & \vdots \\ SAI_{n+1,1} & \dots & SAI_{n+1,m} \end{bmatrix};$$

$$SAI_{ij}\text{: the decriptor for } j^{th} \text{ attribute of } i^{th} \text{Supplier in initiation } i = 1\dots n,\ j = 1\dots m$$

According to the above we have:

$$\text{SRM} = \text{SARM} \S \text{AIM}$$

In which § is as multiplication of values of such matrixes and describing them next to each other.

Taking into consideration the decision variable p about suppliers as follows:

$$\text{Decision Matrix: } DM = \begin{bmatrix} D_1 \\ \vdots \\ D_p \end{bmatrix}; D_l\text{: } l^{th} \text{ Decision Value for } l = 1\dots p$$

SDM matrix are as follows and redefine between suppliers and variables communicate decisions about suppliers:

$$\text{Supplier/Decision Revised Matrix: } SDRM = \begin{bmatrix} SD_{11} & \dots & SD_{1p} \\ \vdots & \ddots & \vdots \\ SD_{n+1,1} & \dots & SD_{n+1,p} \end{bmatrix};$$

$$SD_{il}\text{: the decriptor for } l^{th} \text{ decision of } i^{th} \text{Supplier } i = 1\dots n,\ l = 1\dots p$$

According to the above:

$$\text{SRM} = \text{SDRM} \S \text{DM}$$

In which § is as multiplication of values of such matrixes and describing them next to each other.

If using the information Amendments to the suppliers of the explanatory variables and variables be considered in terms of decision-making in aggregate prosper, we have:

$$\text{SRM} = \text{SRRM} \S \text{M}$$

$$\text{SRRM} = \left\{ \begin{array}{ccc|ccc} SAI_{11} & \dots & SAI_{1m} & SD_{11} & \dots & SD_{1p} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ SAI_{n+1,1} & \dots & SAI_{n+1,m} & SD_{n+1,1} & \dots & SD_{n+1,p} \end{array} \right\}$$

$$M = \left\{ \begin{array}{c} AI_1 \\ \vdots \\ AI_m \\ \hline D_1 \\ \vdots \\ D_p \end{array} \right\};$$

Implementation of CBR Advisor Module

CBR model proposed in this module original cycle is completed. In other words, corrective action and storage of CBR cycle is done in this module.

As seen in Fig. 6, this module receives feedback neural network, as well as the expert Solution Module strategy underlying the proposed CBR, comparing these two definitions on the eve of the similarity of each decision variables and the difference between targets, one of the following approaches taken:

Insert approach: This approach is taken when the proposed solution by the same supplier is certified on select or deselect as well as decision variables like new sample with the most similar example of threshold be more determined. In other words, it confirms Certified Solution proposed by the CBR. In this new approach with the proposed solution as the solution, it is added in the database.

Modify approach: This approach is taken when the proposed solution by the same supplier is certified on select or deselect but variables' similarity of new decision sample be less than the determined threshold. In this new approach with expert opinion as a solution, it is replaced by the most similar case in the case database.

Deleting approach: If the solution proposed by the expert selected or not, is a different supplier selection, this approach is adopted. In this approach the most similar case is removed from the case database.

It is noteworthy to mention that after the election of each of the above-mentioned approaches, it is necessary to update the cluster and rues centers and clustering should also apply modifications.

Proposed Data Mining Revise/Retain Toolkit (PDMRRT2)

For advisor module implement MATLAB software is used. For each of the approaches in this module, special functions were written that according to the proposed approach, expert opinion and threshold of decision variables like one of these functions is executed. Also, since the samples centers to update the database includes numeric variables as well as variables hands are also needed to separate functions for updating values of these variables to be written in cluster centers.

For the first solution, a function is written as addcase.m and for the second function, modify.m. The third strategy was implemented by deleting.m function. In addition to the above measures, cluster members and cluster centers are also updated. What is noteworthy to note that the changes imposed by the expert opinion and add case to the base in the testing phase to a database model to achieve the optimum done and updated expert opinion in validation phase and don't perform.

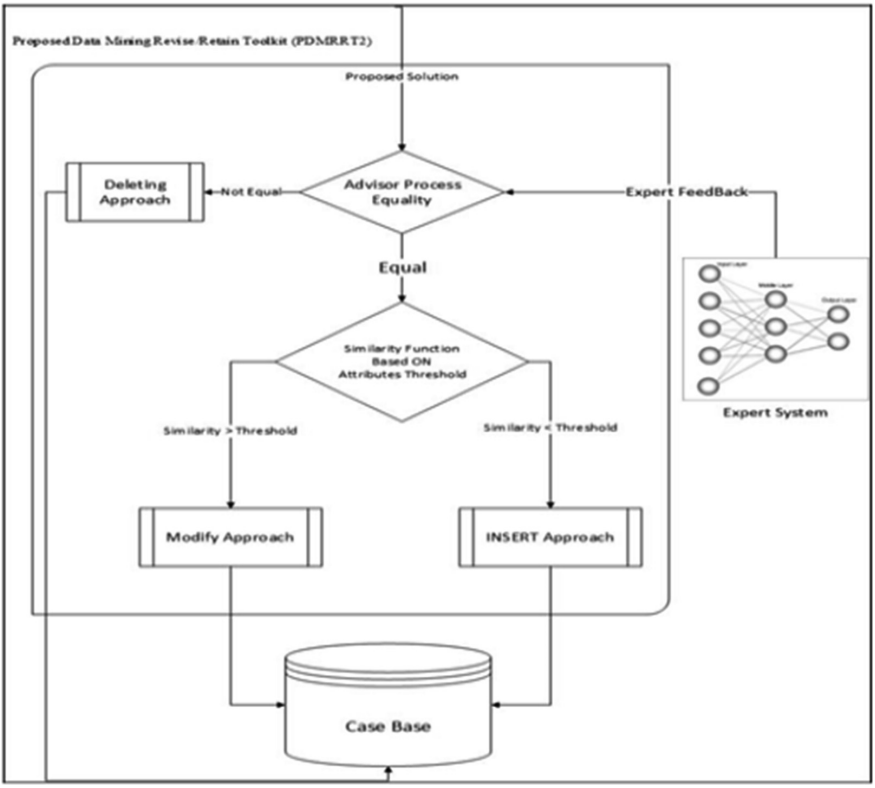

Fig. 9. CBR advisor module architecture.

3 Conclusion

In this paper, a novel model has been proposed based on an object-oriented framework with the supplier selection. This model which leads to optimal selection and ranking of suppliers, reduced the time and cost in the selection process and also reduced human errors by using data mining techniques and neural networks in the reasoning method cycle based on the case. The results of the proposed model are compared with several other models. Results show that using reduced errors, improved accuracy, and efficiency the proposed model has been able to have a good performance in the supplier selection. Also the proposed model has a CBR framework that includes modules for creating and developing the database, the module is responsible for strategy and module amendments which the tools and data mining techniques have been used in each of these modules.

The creation and development of base module about the rules extracted from decision tree trained by the training data set based on the decision variables to develop the database of metadata to suppliers is used. In addition, MLP neural network module is taught as an expert. In the nearest recovery solution for recovery module creates a

new sample and to provide the base were proposed approach to clustering. In clustering, SOM combined technique is used based on K-Prototype and also according to the Davies-Bouldin index, the number of clusters were selected. Finally, the Corrective module according to the proposed strategy, expert opinion and decision variables one of the approaches. Add the similarity threshold, modification and removal is taken. According to the obtained results it can be stated that:

- Reducing the impact of human error by replacing it with expert system based on neural network is another useful outcome of this study to explore it.
- The decision tree is created based on the decision variables based on the weight and characteristics are obtained dependency. As a result, rules of GPA increases the reliability of the base and create more efficiency.

To evaluate the result of the operation model with the aim of practical application in the oil and gas industry is prepared and implemented, the two tangible results of the model include the discovery of an effective decision variable and also retrieve an ideal supplier of the Chartered Company, were examined by the experts.

The results achieved in the first, showing the 8 decision variables extracted from the model to the experts, they number six variables of the decision were approved. Experts have disagreement among them about the remaining variables in the model proposed code status and role in creating rules. With further study on mining legislation have shown that most of the rules of its front have the code status, In Tully Law, the provider has been selected as the winner. The importance of this intention to experts was offered.

References

1. De Boer, L., Labro, E., Morlacchi, P.: A review of methods supporting supplier selection. Eur. J. Purchasing Supply Manage. **7**(2), 75–89 (2001)
2. Hong, G.H., Ha, S.H.: Evaluating supply partner's capability for seasonal products using machine learning techniques. Comput. Ind. Eng. **54**(4), 721–736 (2008)
3. Timmerman, E.: An approach to vendor performance evaluation. IEEE Eng. Manage. Rev. **15**(3), 14–20 (1987)
4. Weber, C.A., Current, J.R., Desai, A.: Non-cooperative negotiation strategies for vendor selection. Eur. J. Oper. Res. **108**(1), 208–223 (1998)
5. Weber, C.A., Desai, A.: Determination of paths to vendor market efficiency using parallel coordinates representation: a negotiation tool for buyers. Eur. J. Oper. Res. **90**(1), 142–155 (1996)
6. Weber, C.A., Current, J., Desai, A.: An optimization approach to determining the number of vendors to employ. Supply Chain Manage. Int. J. **5**(2), 90–98 (2000)
7. Liu, J., Ding, F.Y., Lall, V.: Using data envelopment analysis to compare suppliers for supplier selection and performance improvement. Supply Chain Manage. Int. J. **5**(3), 143–150 (2000)
8. Parthiban, P., Zubar, H.A., Katakar, P.: Vendor selection problem: a multi-criteria approach based on strategic decisions. Int. J. Prod. Res. **51**(5), 1535–1548 (2013)
9. Holt, G.D.: Which contractor selection methodology? Int. J. Project Manage. **16**(3), 153–164 (1998)

10. Lin, R.H., Chuang, C.L., Liou, J.J., Wu, G.D.: An integrated method for finding key suppliers in SCM. Expert Syst. Appl. **36**(3), 6461–6465 (2009)
11. Humphreys, P., McIvor, R., Chan, F.: Using case-based reasoning to evaluate supplier environmental management performance. Expert Syst. Appl. **25**(2), 141–153 (2003)
12. Choy, K.L., Lee, W.B., Lau, H., Lu, D., Lo, V.: Design of an intelligent supplier relationship management system for new product development. Int. J. Comput. Integr. Manuf. **17**(8), 692–715 (2004)
13. Choy, K.L., Lee, W., Lo, V.: Design of a case based intelligent supplier relationship management system—the integration of supplier rating system and product coding system. Expert Syst. Appl. **25**(1), 87–100 (2003)
14. Lee, E.K., Ha, S., Kim, S.K.: Supplier selection and management system considering relationships in supply chain management. IEEE Trans. Eng. Manage. **48**(3), 307–318 (2001)
15. Liu, Y., Yu, F., Su, S.Y., Lam, H.: A cost–benefit evaluation server for decision support in e-business. Decis. Support Syst. **36**(1), 81–97 (2003)
16. Wang, G., Huang, S.H., Dismukes, J.P.: Product-driven supply chain selection using integrated multi-criteria decision-making methodology. Int. J. Prod. Econ. **91**(1), 1–15 (2004)
17. Çebi, F., Bayraktar, D.: An integrated approach for supplier selection. Logistics Inf. Manage. **16**(6), 395–400 (2003)
18. Kilic, H.S.: An integrated approach for supplier selection in multi-item/multi-supplier environment. Appl. Math. Model. **37**(14–15), 7752–7763 (2013)
19. Kumar, S.K., Tiwari, M.K., Babiceanu, R.F.: Minimisation of supply chain cost with embedded risk using computational intelligence approaches. Int. J. Prod. Res. **48**(13), 3717–3739 (2010)
20. Kumar, M., Vrat, P., Shankar, R.: A fuzzy goal programming approach for vendor selection problem in a supply chain. Comput. Ind. Eng. **46**(1), 69–85 (2004)
21. Kumar, M., Vrat, P., Shankar, R.: A fuzzy programming approach for vendor selection problem in a supply chain. Int. J. Prod. Econ. **101**(2), 273–285 (2006)
22. Kumar, A., Jain, V., Kumar, S.: A comprehensive environment friendly approach for supplier selection. Omega **42**(1), 109–123 (2014)
23. Weber, C.A., Current, J.R., Benton, W.C.: Vendor selection criteria and methods. Eur. J. Oper. Res. **50**(1), 2–18 (1991)
24. Das, A., Narasimhan, R., Talluri, S.: Supplier integration—finding an optimal configuration. J. Oper. Manage. **24**(5), 563–582 (2006)
25. Chai, J., Liu, J.N., Ngai, E.W.: Application of decision-making techniques in supplier selection: a systematic review of literature. Expert Syst. Appl. **40**(10), 3872–3885 (2013)
26. Saaty, T.L.: What is the analytic hierarchy process? Springer (1988)
27. De Boer, L., van der Wegen, L., Telgen, J.: Outranking methods in support of supplier selection. Eur. J. Purchasing Supply Manage. **4**(2–3), 109–118 (1998)
28. Roodhooft, F., Konings, J.: Vendor selection and evaluation an activity based costing approach. Eur. J. Oper. Res. **96**(1), 97–102 (1997)
29. Monczka, R.M., Trecha, S.J.: Cost-based supplier performance evaluation. J. Purchasing Mater. Manag. **24**(1), 2–7 (1988)
30. Degraeve, Z., Labro, E., Roodhooft, F.: An evaluation of vendor selection models from a total cost of ownership perspective. Eur. J. Oper. Res. **125**(1), 34–58 (2000)
31. Gaballa, A.A.: Minimum cost allocation of tenders. J. Oper. Res. Soc. **25**(3), 389–398 (1974)
32. Ghodsypour, S.H., O'brien, C.: The total cost of logistics in supplier selection, under conditions of multiple sourcing, multiple criteria and capacity constraint. Int. J. Prod. Econ. **73**(1), 15–27 (2001)

33. Weber, C.A., Current, J.R.: A multiobjective approach to vendor selection. Eur. J. Oper. Res. **68**(2), 173–184 (1993)
34. Zhao, K., Yu, X.: A case based reasoning approach on supplier selection in petroleum enterprises. Expert Syst. Appl. **38**(6), 6839–6847 (2011)
35. Li, H., Sun, J.: Case-based reasoning ensemble and business application: a computational approach from multiple case representations driven by randomness. Expert Syst. Appl. **39**(3), 3298–3310 (2012)
36. Schank, R.C.: Dynamic Memory: A Theory of Reminding and Learning in Computers and People, vol. 240. University Press, Cambridge (1982)
37. Aamodt, A., Sandtorv, H.A., Winnem, O.M.: Combining case based reasoning and data mining-a way of revealing and reusing rams experience. In: Safety and Reliability; Proceedings of ESREL, vol. 98, pp. 16–19 (1998)
38. Li, S., Ragu-Nathan, B., Ragu-Nathan, T.S., Rao, S.S.: The impact of supply chain management practices on competitive advantage and organizational performance. Omega **34**(2), 107–124 (2006)
39. Huang, G., Li, X., He, J., Li, X.: Data mining via minimal spanning tree clustering for prolonging lifetime of wireless sensor networks. Int. J. Inf. Technol. Decis. Making **6**(02), 235–251 (2007)
40. Kim, K.S., Han, I.: The cluster-indexing method for case-based reasoning using self-organizing maps and learning vector quantization for bond rating cases. Expert Syst. Appl. **21**(3), 147–156 (2001)
41. Mosaddar, D., Shojaie, A.A.: A data mining model to identify inefficient maintenance activities. Int. J. Syst. Assur. Eng. Manage. **4**(2), 182–192 (2013)
42. Shearer, C.: The CRISP-DM model: the new blueprint for data mining. J. Data Warehousing **5**(4), 13–22 (2000)